

THE FRUIT PREDICTOR: YOUR FIRST MACHINE LEARNING MODEL

Build a Simple ML Model with Scikit-Learn — Student Handbook & Practical Code Guide

Module: Introduction to Machine Learning

Tool: Python & Scikit-Learn

Role: Vishleshak (Young ML Explorer & Analyst)

1 Introduction: Can You Guess The Mystery Fruit?

Welcome to the world of Artificial Intelligence and Machine Learning! Think like a detective unraveling a mystery. Imagine someone gives you three clues:

The Mystery Fruit Detective Challenge

- **Clue 1 (Texture):** Rough, bumpy skin on the outside.
- **Clue 2 (Structure):** Filled with many seeds inside.
- **Clue 3 (Taste):** Soft, creamy, and wonderfully sweet taste.

What fruit could this be? By observing patterns and connecting clues, your brain makes a smart guess. Machine Learning is about teaching computers to do this exact detective work!

2 Learning from Examples: Alphonso vs. Langra Mangoes

Traditional computer programs follow strict, step-by-step rules written by humans. But Machine Learning (ML) allows computers to learn directly from many real-world examples instead of following rigid rules.

How You Learn vs. How ML Learns: Imagine teaching a friend how to distinguish between an *Alphonso mango* (bright saffron-yellow, sweet aroma, oval shape) and a *Langra mango* (greenish-yellow skin even when ripe, fibrous, distinct citrusy aroma). Rather than explaining complex rules, you show them 50 physical examples of each type! After looking at 100 mangoes, their brain naturally recognizes the subtle patterns. That is precisely how Machine Learning algorithms learn to classify objects!

3 Features: The Clues to Identify Objects

In Machine Learning, we call clues **Features**. Features are measurable characteristics or properties that help a computer tell objects apart.

FEATURE 01: WEIGHT (G)

How heavy is the fruit? Heavier fruits often have more juice or larger seeds (e.g., 200g Mango vs. 120g Banana).

FEATURE 02: TEXTURE

Is the outer skin smooth, rough, bumpy, or leathery? Texture provides distinct surface clues (coded as numbers: 0=Smooth, 1=Rough).

FEATURE 03: COLOR

What color do you see? Green, yellow, orange, or red gives vital hints about fruit type and ripeness (coded: 0=Yellow, 1=Green, 2=Orange).

Indian Market Example: Picking the Sweetest Watermelon

When visiting the vegetable market, experienced buyers select a ripe, sweet watermelon by checking its weight (feeling heavy for its size), its color (dark green with a dull yellow belly patch), and listening to the hollow tapping sound. Computers utilize similar numeric clues (features) to make smart predictions!

4 The 3-Step Machine Learning Cycle

Every machine learning project follows a continuous 3-step pipeline:

Step 1: Train	Step 2: Test	Step 3: Predict
Show the computer many labeled fruit examples containing their features (weight, texture, color) along with their true names (labels).	Present a new, unseen fruit to the trained model to evaluate if it correctly identifies the fruit based on learned patterns.	Deploy the model into real-life applications to autonomously recognize and classify fruits automatically!

Pipeline Formula: Input (Features) → Model (Brain Builder) → Output (Prediction / Anumaan)

5 Meet scikit-learn: The Brain Builder & Decision Trees

Scikit-learn (sklearn) is a popular Python library designed for machine learning. Think of scikit-learn as an inventor's toolbox! We feed feature data into the toolbox, and it constructs a smart Decision Tree brain for us.

What is a Decision Tree? A Decision Tree makes predictions by asking a sequence of simple Yes/No questions, just like playing a game of 20 Questions!

- **Question 1:** Is the fruit weight > 150 grams?
 - If **YES**: Is the texture smooth? → **Mango!**
 - If **NO**: Is the skin texture smooth and color yellow? → **Banana!**
 - If texture is rough & color green: → **Guava!**

6 Hands-On Dataset Walkthrough & Prediction (Anumaan)

Let's examine the exact training dataset used to teach our Fruit Predictor model:

Sample #	Feature 1: Weight (g)	Feature 2: Texture	Feature 3: Color	Target Label (Fruit Name)
Fruit 1	200	Smooth (0)	Yellow (0)	Mango
Fruit 2	120	Smooth (0)	Yellow (0)	Banana
Fruit 3	150	Rough (1)	Green (1)	Guava
Fruit 4	180	Smooth (0)	Orange (2)	Mango
Fruit 5	100	Smooth (0)	Yellow (0)	Banana

Predicting the 6th Mystery Fruit (Anumaan)

Now, a new unknown fruit arrives at your lab table: **Weight = 190g, Texture = Smooth (0), Color = Yellow (0)**. The trained Decision Tree compares these clues with its past experience. Since weight > 150g and texture is smooth, the model makes an *Anumaan* (prediction) with high confidence: **Mango!**

7 Step-by-Step Code Guide: Building the Model with Scikit-Learn

Follow this step-by-step practical guide to write your Python script and build your very first machine learning classifier using scikit-learn.

STEP 1: Import Required Libraries & Prepare the Dataset

In Python, computers understand numbers better than text words. So we convert textures (0 = Smooth, 1 = Rough) and colors (0 = Yellow, 1 = Green, 2 = Orange) into numbers.

```
# Import the Decision Tree Classifier from Scikit-Learn
from sklearn.tree import DecisionTreeClassifier

# Features: [Weight (grams), Texture (0:Smooth, 1:Rough), Color (0:Yellow, 1:Green, 2:Orange)]
X_train = [
    [200, 0, 0], # Fruit 1: 200g, Smooth, Yellow
    [120, 0, 0], # Fruit 2: 120g, Smooth, Yellow
```

```
[150, 1, 1], # Fruit 3: 150g, Rough, Green
[180, 0, 2], # Fruit 4: 180g, Smooth, Orange
[100, 0, 0]  # Fruit 5: 100g, Smooth, Yellow
]

# Target Labels: The true names of the fruits
y_train = ["Mango", "Banana", "Guava", "Mango", "Banana"]
```

STEP 2: Initialize & Train the Decision Tree Model

Next, create an instance of `DecisionTreeClassifier` and call the `.fit()` method to train the model on our data.

```
# Create the Decision Tree Classifier object (The Brain Builder)
model = DecisionTreeClassifier()

# Train (fit) the model with our training dataset
model.fit(X_train, y_train)

print("Model Training Complete! The Decision Tree is ready.")
```

STEP 3: Make Predictions on New Mystery Data (Anumaan)

Use the `.predict()` function to pass a new mystery fruit sample and let the trained model determine its label.

```
# Mystery Fruit: Weight = 190g, Texture = Smooth (0), Color = Yellow (0)
mystery_fruit = [[190, 0, 0]]

# Ask the model to make a prediction
prediction = model.predict(mystery_fruit)

print("Predicted Fruit Name (Anumaan):", prediction[0])
# Output: Predicted Fruit Name (Anumaan): Mango
```

8 Key Takeaways for Young ML Explorers

- **Features are Clues:** Computers cannot see or feel objects like humans do; they rely on numerical and categorical features (weight, texture, color) to understand objects.
- **Training Teaches Patterns:** The more high-quality examples you show to a model during training, the smarter and more accurate its predictions become.
- **Prediction Powers Real Life:** Once trained, models can instantly make predictions (*Anumaan*) on new, unseen data in real-world applications.

9 Extended ML Worksheet & Model Architecture Blueprint

Put on your Vishleshak (Analyst) hat! Complete this worksheet to design your own custom Machine Learning Predictor model.

☐ Project Concept & Problem Definition

1. What real-world items do you want your ML model to predict/classify? (e.g., Types of leaves, street vehicles, sports balls)

Feature Selection (Defining Clues)

List 3 distinct physical features (clues) your model will measure to tell items apart:

- **Feature A:** (e.g., Length, Weight, Speed)
- **Feature B:** (e.g., Color, Surface Texture)
- **Feature C:** (e.g., Number of wheels, Sound level)

Design Your Training Table (Dataset Entry)

Fill in 3 training sample rows for your custom project idea:

Sample	Feature A	Feature B	Feature C	Predicted Class Label
Example 1				
Example 2				
Example 3				

Create a Decision Tree Rule

Write one simple Yes/No question your model's Decision Tree will ask first to split the data:

10 Student Practice Questions

Fruit Predictor ML Knowledge Check - Answer all questions in the space provided.

Q1. Rules vs. Examples

Explain how traditional rule-based computer programming differs from Machine Learning. Use the Alphonso vs. Langra mango example to illustrate your explanation.

Q2. Identifying Features

What is a Feature in Machine Learning? Identify the three specific features used by the Fruit Predictor model to classify fruits in our lesson dataset.

Q3. The 3-Step ML Cycle

Name and describe each of the three steps in the Machine Learning cycle. Explain what happens during the Train phase versus the Predict phase.

Q4. Decision Tree Mechanics & Code

How does a Decision Tree algorithm make decisions? In the code guide provided in Section 7, what method is used to train the model and what method is used to make predictions?

Q5. Dataset Analysis & Prediction (Anumaan)

Suppose a new mystery fruit is introduced to our model with the following clues: Weight = 145g, Texture = Rough (1), Color = Green (1). Based on the 5-fruit training dataset, what prediction (Anumaan) will the model make, and why?

Q6. Impact of Training Data Quality

What happens to a Machine Learning model's accuracy if we train it with only 2 fruit examples instead of 100 fruit examples? Explain why having more high-quality training data makes a model smarter.